MOLE CONCEPT

Units conversion (m)

femto (f) → 10^{-15} $1A^\circ = 10^{-10}$ m.
Pico (P) → 10^{-12}
nano (n) → 10^{-9}
micro (μ) → 10^{-6}
milli (m) → 10^{-3}
centi (c) → 10^{-2}
deci (d) → 10^{-1}
deca (da) → 10
hecto (h) → 10^{2}
kilo (K) → 10^{3}
Mega (M) → 10^{6}
Giga (G) → 10^{9}
Tera (T) → 10^{12}
Peta (P) → 10^{15}

MATTER

① Mixture → More than 1 type of molecule / atom.
ⓐ Homogeneous – sugar soln.
ⓑ Heterogeneous – sugar + salt

② Pure substance → only 1 type of molecule / atom.
ⓐ Element – $H_2(g)$, $O_2(g)$
ⓑ Molecule – H_2O, CO_2 etc.

Calculation of e^-, p^+, n in molecule

neutral atom/molecule ; $n_p = n_e$

Suppose A_xB_y

① total no. of e^- in $A_xB_y = (\sum z_i x_i) \times$ no. of molecules of A_xB_y

② no. of neutron $= (\sum (A-Z)_i x_i) \times$ no. of molecule of A_xB_y

Suppose A_xB_y

① Total no. of atom 'A' $= x \times$ no. of molecule of A_xB_y

② total no. of atom 'B' $= y \times$ no. of molecule (A_xB_y)

③ total no. of atom $= (x+y) \times$ no. of molecule (A_xB_y)

Calculation of e^-, p^+, n in ions

→ only e^- transfer in formation of ions

- no. of $p^+ = (\sum z_i x_i) \times$ no. of ions (cation/anion)
- no. of $n^\circ = (\sum (A-Z)_i x_i) \times$ no. of ion
- no. of e^- of cation $= (\sum z_i x_i - n) \times$ no. of ions
- no. of e^- of anion $= (\sum z_i x_i + n) \times$ no. of ions.

MOLE

$A_xB_y \rightarrow xA + yB$

- no. of moles of A $= x \times$ no. of mole A_xB_y
- no. of moles of B $= y \times$ no. of mole A_xB_y
- ~~no~~ Total moles of atom $= (x+y) \times$ no. of moles A_xB_y

Suppose A_xB_y

- no. of 'A' atom $= x \times$ moles of $A_xB_y \times N_A$
- no. of molecule of A_xB_y = moles $(A_xB_y) \times N_A$
- no. of total atom $= (x+y) \times$ moles of $A_xB_y \times N_A$.

Calculation of moles

$$\boxed{n = \frac{\text{given mass}}{\text{Molar mass}} = \frac{\text{given number}}{6.022 \times 10^{23}\,(N_A)}}$$

↳ Applicable in all physical state.

→ Mass of 1 mole = M.wt in gram

→ Mass of substance = no. of mole × M.wt

1 mole of Ideal gas = 22.4 L at STP.

$$\boxed{n = \frac{\text{Vol. of gas at STP}}{22.4}}$$ → only applicable at STP i.e 273 K 1 atm.

$$\boxed{n = \frac{PV}{RT}}$$ → only Applicable for gas. (At any Temp & pressure).

H_2O

$$\boxed{\text{density} = \frac{M}{V} = \frac{1g}{ML}}$$

Bpt = 100°c = 373 K

100°c ≤ t → H_2O (vap)

100°c > t → H_2O (L)

ATOM

→ Smallest unit of Matter.

① Atomic mass = mass number = Mass of [proton + neutron]

Mass of a proton $= 1.67 \times 10^{-24}$ g = 1 amu

1 amu $= \frac{1}{12}$th of Mass of C-12 atom

1 amu $= 1.67 \times 10^{-24}$ g $= \frac{1}{N_A}$

→ Atomic Mass = 3 types

① Actual atomic Mass [AAM]

② Relative atomic Mass [RAM]

③ Gram atomic Mass [GAM] or Molar mass

② AAM = mass no. $\times 1.67 \times 10^{-24}$ g

RAM = Mass no. (amu)

GAM = Mass no. (gm) = AAM × NA

③ mass in amu = atoms

Mass in gm = mol atoms

④ 1 mol atom = 1 g atom

1 mol molecule = 1 g molecule

Molecular Mass

Avg. Molecular wt $= \frac{\Sigma M_i n_i}{\Sigma n_i} = \frac{M_1 n_1 + M_2 n_2}{n_1 + n_2}$

① Mass of 1 molecule = (molecular mass) amu

② Mass of 1 mole of molecule = (molecular mass) gram.

③ 22.4 L at S.T.P (gas) = (M.wt) gm.

④ 6.022×10^{23} no. of molecule = (M.wt) gm

Average Atomic Mass

① $A_{avg} = \frac{\Sigma A_i n_i}{\Sigma n_i}$

② $A_{min} < A_{avg} < A_{max}$

③ Aavg. atomic mass, jyada wale % ke pass hota hain.

vapour density

① V.D of gas $= \frac{\text{M.wt of gas}}{2}$

② $(V.D)_{avg} = \left(\frac{M.wt}{2}\right)_{Avg}$ (gas mix)

Mass % of element $= \frac{\text{total mass of element}}{\text{M.wt of compound}} \times 100$

→ % dekar kuch v puchhe to lga do.

Empirical formula

→ Minimum simple molar Mass

→ should be whole no. & Minimum

Molecular formula = n × Empirical formula

Concentration

⇒ Mass % → Amount of solute in 100 g solution

Mass % of solute $= \frac{w_1}{w_1 + w_2} \times 100$

Mass % of solvent $= \frac{w_2}{w_1 + w_2} \times 100$

⇒ parts per million (ppm)

$PPM = \frac{w_1}{w_1 + w_2} \times 10^6$

ppm = mass % $\times 10^4$

⇒ parts per billion (PPB)

$PPB = \frac{w_1}{w_1 + w_2} \times 10^9$

PPB = mass % $\times 10^7$

PPB = PPM $\times 10^3$

$x\%(w/w) + y\%(w/w)$

Mass of resultant solⁿ $= \left(\frac{x+y}{2}\right)\%$

Note :- Mass of solⁿ alag se given nahi hona chahiye.

$x\%(w/w)$ of solⁿ

Mass of solute $= x$ gram

Mass of solution $= 100$ gram

Mass of solvent $= (100 - x)$ gram

$x\%(w/v)$ of solution

Mass of solute $= x$ gram

~~Mass of solution = 100 gram~~

Volume of solution $= 100$ ml

$$\boxed{x\% = \frac{w}{v} \times 100}$$

$d_{sol^n} = g/ml$ given

$$\boxed{M_{sol^n} = d_{sol^n} \times V_{sol^n}}$$

$x\%(v/v)$ of solution

x ml solute, 100 ml solution

$$\boxed{x\% = \frac{V_1}{V_1+V_2} \times 100}$$

Mole fraction

→ Mole fraction of solute :-

$$\boxed{x_1 = \frac{n_1}{n_1+n_2}} \quad \boxed{x_2 = \frac{n_2}{n_1+n_2}}$$

→ Mole % of solute $= \frac{n_1}{n_1+n_2} \times 100$

$$\boxed{x_1 + x_2 = 1} \quad \boxed{\frac{n_1}{n_2} = \frac{x_1}{x_2}}$$

MOLALITY (m)

$$\boxed{\text{Mass of solvent} = \text{Mass of sol}^n - \text{Mass of solute}}$$

$$m = \frac{n \times 10^3}{\text{mass of solvent (g)}}$$

$m = $ Mole/kg

Representation ⇒ 2 mole kg^{-1} = 2m.

→ If density given

$$\boxed{m = \frac{n \times 10^3}{d^{sol^n} \times Vol^{sol^n} - \text{Mass of solute}}}$$

Relation b/w Mole fraction & Molality

$$\boxed{\frac{x_1}{x_2} = \frac{m \times M.wt \text{ of solvent}}{1000}}$$

Molarity (M)

$$\boxed{M = \frac{(\text{Mole})_{solute}}{\text{Vol. of sol}^n (L)} = \frac{(\text{Mole})_{solute} \times 10^3}{\text{Vol. of sol}^n (ml)}}$$

Note → If solute = Solid

Vol. of solvent = Vol. of solution

$$\boxed{M = \frac{(\text{Mole})_{solute} \times 10^3 \times d_{sol^n}}{(\text{mass})_{sol^n} \rightarrow \text{gram}}}$$

→ Suppose, Molarity of $A_xB_y = M$

Molarity of $A^{+y} = x \times$ molarity $(A_xB_y) = x \times M$

Molarity of $B^{-x} = y \times$ Molarity $(A_xB_y) = y \times M$

Relation b/w Molarity (M) and Molality (m) :-

$$\boxed{m = \frac{M \times 10^3}{d \times 1000 - M \times M.wt \text{ of solute}}}$$

Note :- Above formula used, only vol. of solⁿ consider = 1L = 1000ml.

⇒ Given that,

(i) Mass % $= x\%$

(ii) density of solⁿ $= d$ g/L

$$\boxed{\text{Molarity} = \frac{x \times \text{density of sol}^n}{(M.wt)_{solute} \times 100}}$$

$$\Rightarrow \boxed{\text{No. of mole of solute} = M \times V(\text{Litre})}$$

⇒ Given that, $x\%(w/v)$ of solⁿ.

$$\boxed{\text{Molarity} = \frac{x \times 1000}{(M.wt)_{solute} \times 100} = \frac{x \times 10}{(M.wt)_{solute}}}$$

Molarity of Mix of 2 acid or 2 Base

$$\boxed{M_1V_1 + M_2V_2 = M(V_1+V_2)}$$

Molarity of soln on dilution

$$\boxed{M_1V_1 = M(V_1+V_2)}$$

Stoichiometric calculation

Chem. rkn → $n_1 A \longrightarrow n_2 B + n_3 C$

① Rxn balance honi chahiye

② Given info → convert into Mole.

③ $\frac{(\text{mole})_A}{n_1} = \frac{(\text{mole})_B}{n_2} = \frac{(\text{mole})_C}{n_3}$

Percentage Purity

$$\%\ \text{purity of sample} = \frac{\text{mass of pure comp.}}{\text{mass of sample}} \times 100$$

Findout (mass of pure comp.)

↳ given (mass of sample)

Percentage Yielde

$$\boxed{\%\ \text{Yielde} = \frac{\text{mass of comp. given}}{\text{Mass of comp. calculated}} \times 100}$$

NOTE:- Jis molecule or comp. ke mass question me diya ho, usi ka mass nikal dena.

⇒ Reactant ke respect Me

$$\boxed{\%\ \text{Yielde} = \frac{\text{Mass of reactant calculated}}{\text{Mass of reactn given}} \times 100}$$

Limiting Reagent

① Jab 2-reactant ki value given ho.

② check kaise krte hai → Jiska $\left(\frac{\text{Mole}}{\text{Coefficient}}\right)$ ratio, small hota hai, wahi limiting reagent hota hain.

③ Limiting reagent → Jo reaction Me pahle Khatam hota hain.

④ Jo v nikalna ho, uske $\left(\frac{\text{Mole}}{\text{coeff.}}\right)$, L.R ka $\left(\frac{\text{Mole}}{\text{Coef.}}\right)$ ke equal kr denge.

$$\boxed{\left(\frac{\text{Mole}}{\text{coeff.}}\right)_{L.R} = \left(\frac{\text{Mole}}{\text{Coef.}}\right)_{\text{finding value}}}$$

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